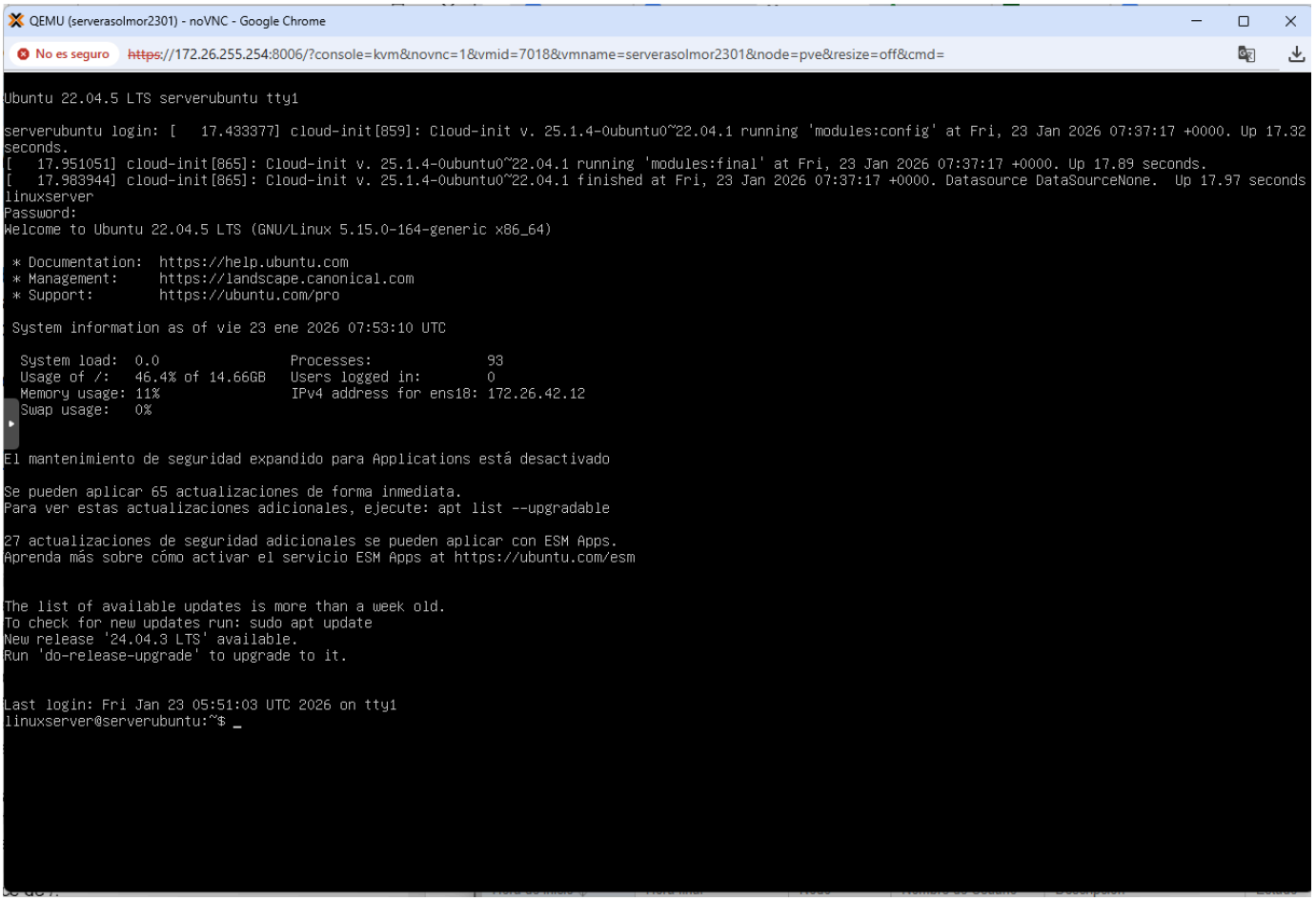
Andrea Solís Moreno 1ºDAW
23/01/2026
SISTEMAS INFORMÁTICOS

PRUEBA PRÁCTICA UNIDAD 4: GESTIÓN DE ARCHIVOS LINUX

Entramos en la consola, iniciamos sesión con el login que es linuxserver y a continuación nos pedirá la contraseña, que es hortensia.



Nos bajamos el siguiente libro de la url con wget.

```
root@serverubuntu:/home# wget https://www.gutenberg.org/files/1259/1259-0.txt
```

Al ejecutarlo nos aparece lo siguiente:

```
root@serverubuntu:/home# wget https://www.gutenberg.org/files/1259/1259-0.txt
--2026-01-23 07:57:50--  https://www.gutenberg.org/files/1259/1259-0.txt
Resolving www.gutenberg.org (www.gutenberg.org)... 152.19.134.47, 2610:28:3090:3000:0:bad:cafe:47
Connecting to www.gutenberg.org (www.gutenberg.org)|152.19.134.47|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1420236 (1,4M) [text/plain]
Saving to: '1259-0.txt'

1259-0.txt                          100%[=================================================================

2026-01-23 07:57:52 (1,90 MB/s) - '1259-0.txt' saved [1420236/1420236]

root@serverubuntu:/home# _
```

1.Renombrar el fichero descargado como libro1.txt

Para ello uso el comando mv → mv 1259-0.txt libro1.txt

```
root@serverubuntu:/home# mv 1259-0.txt libro1.txt
root@serverubuntu:/home#
```

Vamos a comprobarlo con el comando ls:

```
root@serverubuntu:/home# mv 1259-0.txt libro1.txt
root@serverubuntu:/home# ls
andreasm  comando-grep  libro1.txt  linuxserver  Prueba-sistemas  Sistemas_Informaticos
root@serverubuntu:/home# _
```

2.Mostrar todas las líneas que comiencen por Cha.

Antes de ello voy a ejecutar un comando para evitar saltos de líneas con el comando →
sed -i 's/\r//' libro1.txt

```
root@serverubuntu:/home# sed -i 's/\r//' libro1.txt
root@serverubuntu:/home# _
```

Ahora es cuando uso el comando → grep "^Cha" libro1.txt

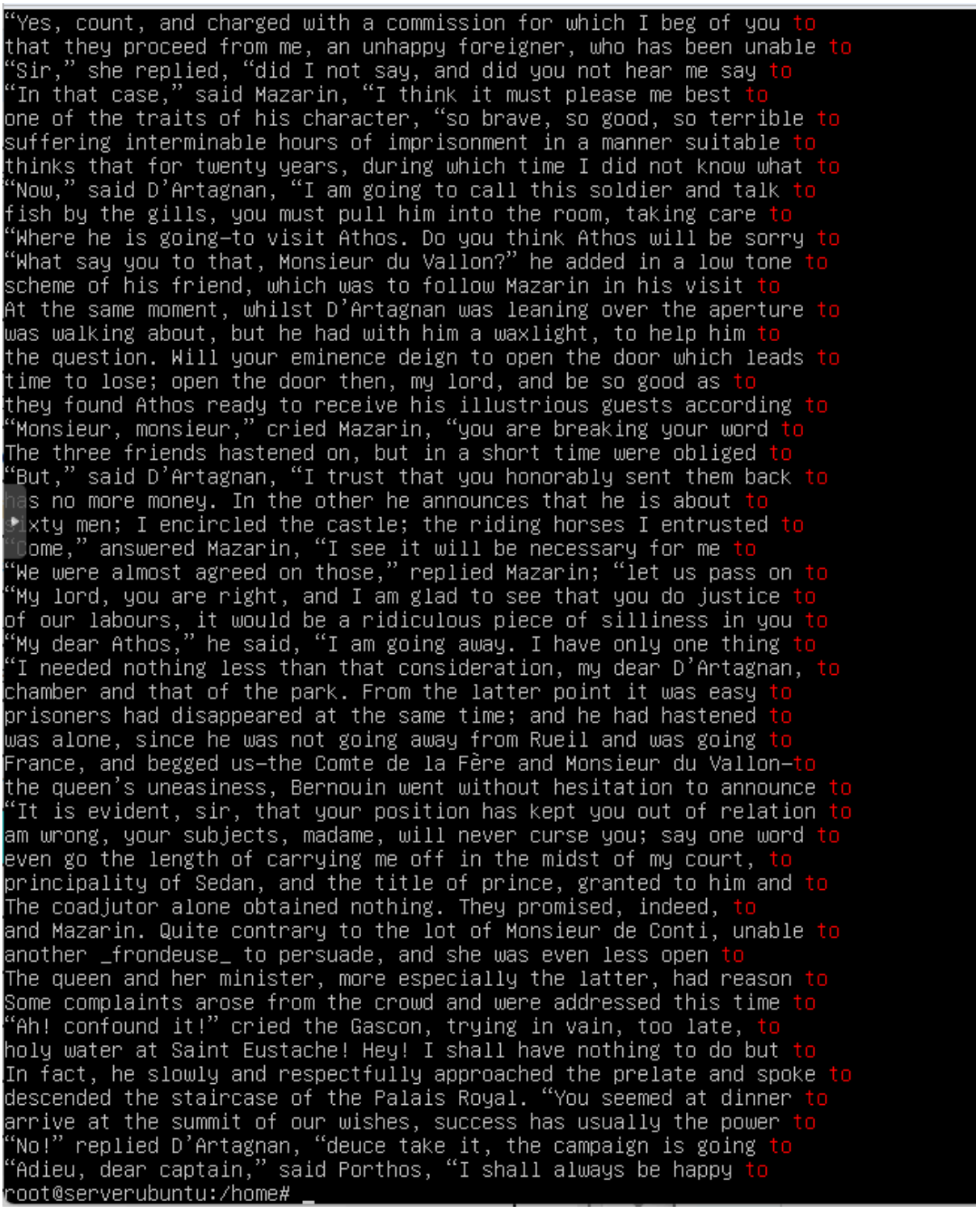
```
root@serverubuntu:/home# grep "^Cha" libro1.txt
```

```
Charles listened with marked attention, passing over the insults,
Charles also had recognized the four Frenchmen. He put one hand on his
Charles rose and turned toward Parry, whom he saw pale and with his
Charles wiped his face and said with a sad smile: “Poor wretch, for
Chapter LXIV.
Charles returned in silence to his chair, rested his elbow on the
Charles clasped his hands, amazed and stupefied to find that these
Charles shook his head.
Chapter LXV.
Charles was impatiently awaiting his return. At the moment he
Charles evinced, as he learned these tidings. But Aramis was livid. His
Charles perceived it and took his hand.
Chapter LXVI.
Charles’s heart melted within him at the sight of those two children,
Charles, weary of waiting for death, had suddenly decided to go to meet
Charles accompanied these words with a look designed to penetrate the
Chapter LXVII.
Chapter LXVIII.
Chapter LXIX.
Chapter LXX.
arles. Now, not a second to lose, for a second lost will perhaps save
apter LXXI.
apter LXXII.
Chapter LXIII.
Charles! incendiary! I recommend you to sink forthwith to the bottom of
Chapter LXIV.
Charles; because, instead of helping cut off the royal head condemned
Chapter LXXV.
Chancellor Séguier, implying that if the parliament did not humble
Chapter LXXVI.
Charles I. before his death had placed in the hands of Aramis. Since
Charenton?”
Chapter LXXVII.
Chapter LXXVIII.
Chanleu, whose fire at one time repulsed the royal regiment, thought
Chapter LXXIX.
Chapter LXXX.
Chapter LXXXI.
Chapter LXXXII.
Chapter LXXXIII.
Chapter LXXXIV.
Chapter LXXXV.
Chapter LXXXVI.
Chapter LXXXVII.
Charenton, will be near Compiègne or Villars-Cotterets.”
Chapter LXXXVIII.
Charles I., brother-in-law of the late king, your husband, the consort
Chapter LXXXIX.
Chapter XC.
root@serverubuntu:/home# _
```

3.Mostrar todas las líneas que terminen en to.

Para ello uso el comando → grep “to$” libro1.txt

```
root@serverubuntu:/home# grep "to$" libro1.txt
```



4.Mostrar todas las líneas que contengan la palabra madame.

Uso el comando→ grep “madame” libro1.txt



```
“Every one, madame.”
madame; believe me, I know the people; they are like a great child, who
“I shall tell them, madame, and those who accompany me will say the
“Well, madame,” replied the valet, “I followed them as far as the
“Yes, madame,” said D’Artagnan; “but get in fearlessly, for I myself
“How, madame!” said the valet, “in the same carriage as your
“Yes, madame,” Châtillon stammered out.
“Just so madame,” replied Châtillon, in so low a voice that though the
“Now, madame, nothing remains to me except to proffer my respectful
“I beg you to excuse me, madame; I wish to add nothing to the recital
“Yes, madame, another service,” said Athos, shocked that the queen did
“Yes, madame, here I have a right to do so, for it concerns Monsieur
“I wish to do, madame, all that your majesty pleases.”
“I have another name, madame—I am called Athos.”
“Yes, madame, I expected nothing less from the justice of your majesty.
“To promote your majesty’s welfare, madame,” replied Mazarin, fixing
“I am thinking, madame, of the allusion made by that insolent
not, madame? I shall tell him, therefore, and in doing so I shall
France? Thrown over the foot of the royal bed, my simar, madame, looks
“Pardon, madame, but I think that either I have expressed myself badly
dame, it proved to be as we had supposed; there was a mistake.
our pardon, madame, I know all that. And therefore my friends will
m obliged to do so. Believe me, madame, as true a thing as it is that
madame, no compassion for your servants who for twenty years have
me, madame—at me, whom you accuse of speaking loud and threateningly.
am wrong, your subjects, madame, will never curse you; say one word to
“Because, madame, it was necessary to prove to your majesty one thing
“We have not thought about it _yet_, madame,” answered D’Artagnan, with
“Alas, madame!” exclaimed D’Artagnan, “this proves to me that not till
“We do ourselves justice, madame, even in assuming the high position
“I believe you, madame, to be a great and powerful queen, and that it
“The place, madame, is vacant, and although ’tis a year since Monsieur
like me, madame; he occupied that post for twenty years.”
“Undoubtedly, madame,” said D’Artagnan, taking the document and bowing,
“No, madame, something easier to grant.”
“Monsieur de Longueville is of royal blood, madame,” said D’Artagnan.
“His son, madame, must be, since the husband of the son’s mother is.”
“No, madame, for I presume that the king, standing godfather to him,
“We shall search for some, madame, and I think we can find a little,
“What next, madame?”
“Yes, madame, the Comte de la Fère.”
“There is the Comte de la Fère, madame. The Comte de la Fère is not a
have time to sign to-morrow. Consent, then, I beg you, madame, to write
de Mazarin to your majesty without conditions. Behold, madame! here are
“Yes, madame; I have that of the Garter, which King Charles I. gave me
“No, madame; if you have forgotten the history of Œdipus, I, at least,
“You are adorable, madame.”
“Alas, madame!” exclaimed Athos, “to-day love is like war—the
“Reassure yourself, madame, I will answer for the king’s safety.”
root@serverubuntu:/home#
```

5.Mostrar todas las líneas que comiencen por una letra mayúscula.

minúscula

Uso el comando → grep “^[A-Z]” libro1.txt

```
root@serverubuntu:/home# grep "^[A-Z]" libro1.txt
```

Comminges returned to his post. An effort was made and the living
Some complaints arose from the crowd and were addressed this time to
But as if the multitude had waited only for this demonstration to burst
At the same time through the streets of Grenelle, Saint Honoré, and Du
Coq, a double stream of people broke the feeble hedge of Swiss guards
This new eruption was more dangerous than the others, being composed of
Each of these mobs was led by a chief, one of whom appeared to belong,
There was a shock which was perceived even in the royal carriage.
Myriads of hoarse cries, forming one vast uproar, were heard, mingled
The escort divided into two files. One of them passed around to the
D'Artagnan, the other Porthos. Then came a skirmish, the more terrible
D'Artagnan. And turning with that fury which made him so formidable, he
At these words the man with a pistol and sword raised his head, but it
Rochefort extended his hand to D'Artagnan, who took it. The count
Indeed, as if De Rochefort had been the very soul of the attack, the
Coq, and the portion of the mob he assailed disappeared like smoke,
D'Artagnan returned to help Porthos, if Porthos needed help; but
Porthos, for his part, had done his work as conscientiously as
D'Artagnan. The left of the carriage was as well cleared as the right,
Porthos looked very melancholy.
 fact, the procession arrived, in safety at Notre Dame, at the front
 the service was drawing to a close a boy entered the church in great
zin felt some one pulling his sleeve. He lowered to earth his eyes,
Italian to the gallows!' Isn't that what you would have me shout?"
Say, then, Father Bazin—the heir of Monsieur Maillard, the giver of
In fact, he slowly and respectfully approached the prelate and spoke to
When the _Te Deum_ was over, the coadjutor, without stopping to change
A moment later Friquet went out, carrying in his hand a large leather
He could say no more; but though he hadn't strength to speak he had
Mother Nanette began by picking up the louis; then she picked up
Friquet.
In the meantime the cortège returned to the Palais Royal.
Treat him kindly, therefore, in the future."
That was a great honor for D'Artagnan and for Porthos. Porthos was
Chapter XC.
Conclusion.
On going home, the two friends found a letter from Athos, who desired
The friends went to bed early, but neither of them slept. When we
The next day at the appointed hour they went to see Athos and found him
Noisy-le-Sec."
D'Artagnan, I am nothing more than a good honest country gentleman.
Raoul has no fortune other than I possess, poor child! and I must take
The four friends embraced with tears in their eyes.
Then they departed, without knowing whether they would ever see each
D'Artagnan returned to the Rue Tiquetonne with Porthos, still possessed
D'Artagnan's Gasconades.
The two friends, after a cordial pressure of the hands, separated.
D'Artagnan was standing at the door looking after Porthos with a
And Porthos, enchanted at having relieved his mind, rejoined Mousqueton
root@serverubuntu:/home# _

6. Mostrar todas las líneas que contengan números.

Uso el comando → grep “^[0-9]” libro1.txt

```
root@serverubuntu:/home# grep "^[0-9]" libro1.txt
```

```
root@serverubuntu:/home# grep "^[0-9]" libro1.txt
1       The Three Musketeers        1257        1625-1628    1
2       Twenty Years After          1259        1648-1649    2
3       The Vicomte de Bragelonne   2609        1660         3         1-75
4       Ten Years Later             2681        1660-1661    3         76-140
5       Louise de la Vallière       2710        1661         3         141-208
6       The Man in the Iron Mask    2759        1661-1673    3         209-269
2681]
12 Rue des Fossoyeurs. D'Artagnan had not failed in his career, but
root@serverubuntu:/home# _
```

7. Mostrar todas las líneas que acaben en punto.

Uso el comando → grep “.$” libro1.txt

```
root@serverubuntu:/home# grep ".$" libro1.txt
```

```
me.
The four friends embraced with tears in their eyes.
Then they departed, without knowing whether they would ever see
other again.
D'Artagnan returned to the Rue Tiquetonne with Porthos, still p
by the wish to find out who the man was that he had killed. On
at the Hôtel de la Chevrette they found the baron's equipage al
and Mousqueton on his saddle.
```

8.Contar las líneas que contienen la palabra minister.

Uso el comando → grep -o “minister” libro1.txt | wc -1

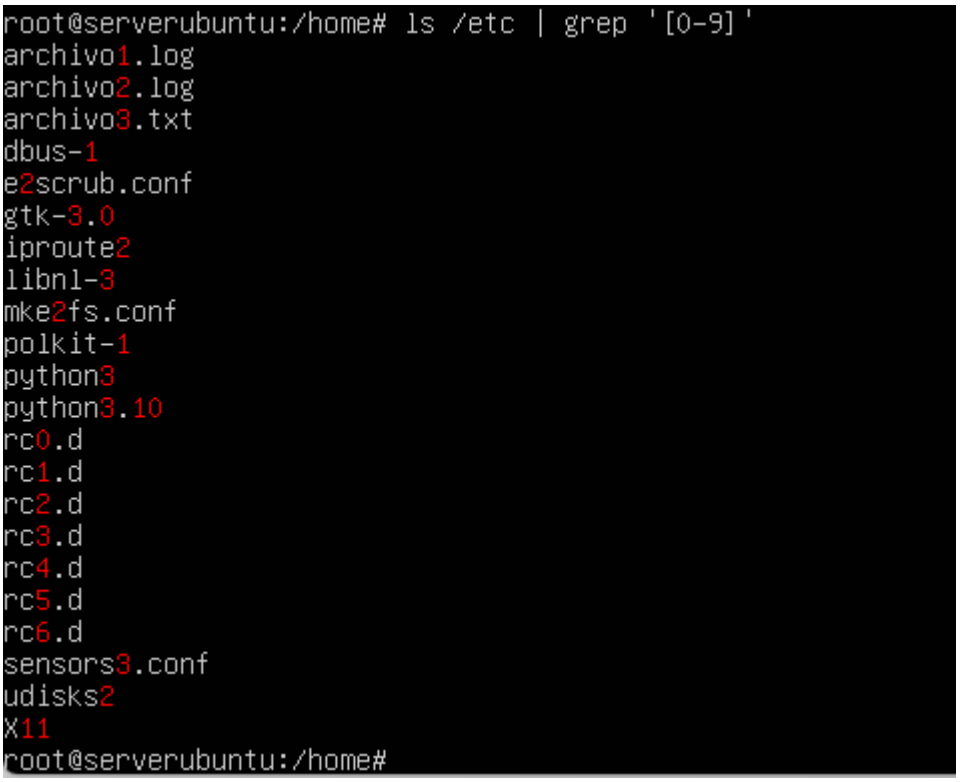
```
root@serverubuntu:/home# grep -o "minister" libro1.txt | wc -l
```

```
root@serverubuntu:/home# grep -o "minister" libro1.txt | wc -l
78
root@serverubuntu:/home#
```

9. Listar los directorios de /etc que contengan un número en su nombre.

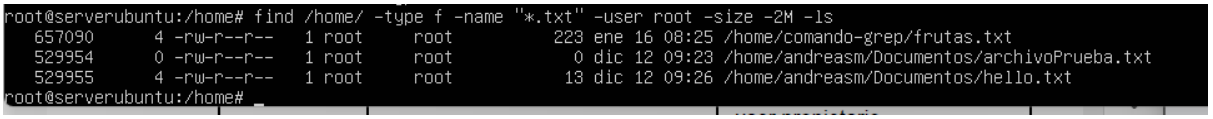
Uso el comando → ls /etc | grep '[0-9]'





10.Encontrar todos los ficheros con extensión txt, propiedad del usuario root, con tamaño inferior a 2 megas.

Uso el comando → find



11.Mostrar todos los usuarios registrados en el sistema ordenados en orden alfabético.

Uso el comando→ cut -f1 -d: /etc/passwd | wc -l

```
root@serverubuntu:/home# cut -f1 -d: /etc/passwd | wc -l
34
root@serverubuntu:/home# _
```

12. Mostrar todos los usuarios del sistema donde el shell que los atiende sea bash.

Uso el comando → grep '/bin/bash' /etc/passwd | cut -d: -f1

```
root@serverubuntu:/home# grep '/bin/bash' /etc/passwd | cut -d: -f1_
```

```
root@serverubuntu:/home# grep '/bin/bash' /etc/passwd | cut -d: -f1
root
linuxserver
root@serverubuntu:/home#
```

13. Encontrar todos los ficheros con extension ko que se encuentren dentro de /lib/modules.

Uso el comando → find /lib/modules "*.ko"

```
root@serverubuntu:/home# find /lib/modules "*.ko"
```



14. Mostrar una lista de ficheros enlace de /.

Uso el comando → ls -la /

```
root@serverubuntu:/home# ls -la/
ls: invalid option -- '/'
Try 'ls --help' for more information.
root@serverubuntu:/home# ls -la /
total 2097236
drwxr-xr-x  21 root root       4096 ene 16 17:47 .
drwxr-xr-x  21 root root       4096 ene 16 17:47 ..
lrwxrwxrwx   1 root root          7 sep 11  2024 bin -> usr/bin
drwxr-xr-x   4 root root       4096 ene  9 08:37 boot
dr-xr-xr-x   2 root root       4096 sep 11  2024 cdrom
drwxr-xr-x  20 root root       4040 ene 23 07:37 dev
drwxr-xr-x 107 root root       4096 ene  9 09:03 etc
drwxr-xr-x   7 root root       4096 ene 23 08:07 home
lrwxrwxrwx   1 root root          7 sep 11  2024 lib -> usr/lib
lrwxrwxrwx   1 root root          9 sep 11  2024 lib32 -> usr/lib32
lrwxrwxrwx   1 root root          9 sep 11  2024 lib64 -> usr/lib64
lrwxrwxrwx   1 root root         10 sep 11  2024 libx32 -> usr/libx32
drwx------   2 root root      16384 sep 23 16:12 lost+found
drwxr-xr-x   2 root root       4096 sep 11  2024 media
drwxr-xr-x   2 root root       4096 sep 11  2024 mnt
drwxr-xr-x   2 root root       4096 sep 11  2024 opt
dr-xr-xr-x 156 root root          0 ene 23 07:37 proc
lrwxrwxrwx   1 root root         21 ene 16 17:47 Prueba-sistemas -> /home/Prueba-sistemas
drwx------   5 root root       4096 ene  9 09:00 root
drwxr-xr-x  29 root root        860 ene 23 07:53 run
lrwxrwxrwx   1 root root          8 sep 11  2024 sbin -> usr/sbin
drwxr-xr-x   2 root root       4096 ene  9 09:21 SIST_INF
drwxr-xr-x   6 root root       4096 sep 11  2024 snap
drwxr-xr-x   2 root root       4096 sep 11  2024 srv
-rw-------   1 root root 2147483648 sep 23 16:14 swap.img
dr-xr-xr-x  13 root root          0 ene 23 07:37 sys
drwxrwxrwt  14 root root       4096 ene 23 07:42 tmp
drwxr-xr-x  14 root root       4096 sep 11  2024 usr
drwxr-xr-x  13 root root       4096 sep 11  2024 var
root@serverubuntu:/home#
```

15. Mostrar una lista de ficheros con extension conf de /etc .

Uso el comando → ls /etc/*.conf

```
root@serverubuntu:/home# ls /etc/ *.conf
ls: cannot access '*.conf': No such file or directory
/etc/:
adduser.conf                  dbus-1          host.conf       lsb-release         odbc.ini        rcS.d           sysctl.d
alternatives                  dconf           hostname        lvm                 odbcinst.ini    repaso.txt      systemd
apparmor                      debconf.conf    hosts           machine-id          openal          resolv.conf     terminfo
apparmor.d                    debian_version  hosts.allow     magic               opt             rmt             thermald
apport                        default         hosts.deny      magic.mime          os-release      rpc             timezone
apt                           deluser.conf    init.d          manpath.config      overlayroot.conf rsyslog.conf   tmpfiles.d
archivo1.log                  depmod.d        initramfs-tools mdadm               PackageKit      rsyslog.d       ubuntu-advantage
archivo2.log                  dhcp            inputrc         mime.types          pam.conf        screenrc        ucf.conf
archivo3.txt                  dpkg            iproute2        mke2fs.conf         pam.d           security        udev
archivo.txt                   e2scrub.conf    iscsi           ModemManager        passwd          selinux         udisks2
bash.bashrc                   environment     issue           modprobe.d          passwd-         sensors3.conf   ufw
bash_completion               environment.d   issue.net       modules             perl            sensors.d       update-manager
bash_completion.d             ethertypes      kernel          modules-load.d      pki             services        update-motd.d
bindresvport.blacklist        fonts           landscape       mtab                pm              shadow          update-notifier
binfmt.d                      fstab           ldap            multipath           polkit-1        shadow-         UPower
byobu                         fuse.conf       ld.so.cache     multipath.conf      pollinate       shells          usb_modeswitch.conf
ca-certificates               fwupd           ld.so.conf      mysql               profile         skel            usb_modeswitch.d
ca-certificates.conf          gai.conf        ld.so.conf.d    nanorc              profile.d       sos             vdpau_wrapper.cfg
ca-certificates.conf.dpkg-old ghostscript     legal           needrestart         protocols       ssh             vim
cloud                         glvnd           libaudit.conf   netconfig           pulse           ssl             vmware-tools
console-setup                 groff           libblockdev     netplan             python3         subgid          vtrgb
cron.d                        group           libnl-3         network             python3.10      subgid-         wgetrc
cron.daily                    group-          locale.alias    networkd-dispatcher rc0.d           subuid          X11
cron.hourly                   grub.d          locale.gen      networks            rc1.d           subuid-         xattr.conf
cron.monthly                  gshadow         localtime       newt                rc2.d           sudo.conf       xdg
crontab                       gshadow-        logcheck        nftables.conf       rc3.d           sudoers         zsh_command_not_found
cron.weekly                   gss             login.defs      nsswitch.conf       rc4.d           sudoers.d
cryptsetup-initramfs          gtk-3.0         logrotate.conf  nuevo_archivo.txt   rc5.d           sudo_logsrvd.conf
crypttab                      hdparm.conf     logrotate.d     ODBCDataSources     rc6.d           sysctl.conf
root@serverubuntu:/home#
```

16. Extraer del fichero /etc/passwd los usuarios cuyo nombre comience por las letras "a hasta g".

Uso el comando → awk -F: '$1 ~ /^[a-g]/ {print $1}' /etc/passwd

```
root@serverubuntu:/home# awk -F: '$1 ~/^[a-g]/ {print $1}' /etc/passwd
daemon
bin
games
backup
gnats
fwupd-refresh
root@serverubuntu:/home#
```

17. Mostrar una lista de ficheros de acceso a dispositivo de /dev que sean de tipo bloque.

Uso el comando → ls -l /dev/ | grep '^b'

```
root@serverubuntu:/home# ls -l /dev/ | grep '^b'
brw-rw----  1 root        disk    253,   0 ene 23 07:37 dm-0
brw-rw----  1 root        disk      7,   0 ene 23 07:37 loop0
brw-rw----  1 root        disk      7,   1 ene 23 07:37 loop1
brw-rw----  1 root        disk      7,   2 ene 23 07:37 loop2
brw-rw----  1 root        disk      7,   3 ene 23 07:37 loop3
brw-rw----  1 root        disk      7,   4 ene 23 07:37 loop4
brw-rw----  1 root        disk      7,   5 ene 23 07:37 loop5
brw-rw----  1 root        disk      7,   6 ene 23 07:37 loop6
brw-rw----  1 root        disk      7,   7 ene 23 07:37 loop7
brw-rw----  1 root        disk      8,   0 ene 23 07:37 sda
brw-rw----  1 root        disk      8,   1 ene 23 07:37 sda1
brw-rw----  1 root        disk      8,   2 ene 23 07:37 sda2
brw-rw----  1 root        disk      8,   3 ene 23 07:37 sda3
brw-rw----+ 1 root        cdrom    11,   0 ene 23 07:37 sr0
root@serverubuntu:/home#
```

18. Listar todos los ficheros ordinarios de /var.

Uso el comando → find /var -type f

```
/var/log/dmesg.3.gz
/var/log/bootstrap.log
/var/log/auth.log
/var/log/dmesg.2.gz
/var/log/dmesg
/var/log/fontconfig.log
/var/log/ubuntu-advantage.log
/var/log/cloud-init.log
/var/log/faillog
/var/log/alternatives.log.1
/var/log/syslog.3.gz
/var/log/auth.log.2.gz
/var/log/installer/device-map.json
/var/log/installer/subiquity-server-info.log.1693
/var/log/installer/curtin-install/subiquity-extract.conf
/var/log/installer/curtin-install/subiquity-curtin-apt.conf
/var/log/installer/curtin-install/subiquity-curthooks.conf
/var/log/installer/curtin-install/subiquity-initial.conf
/var/log/installer/curtin-install/subiquity-partitioning.conf
/var/log/installer/curtin-install.log
/var/log/installer/cloud-init-output.log
/var/log/installer/block/discover.log
/var/log/installer/block/probe-data.json
/var/log/installer/subiquity-client-debug.log.1675
/var/log/installer/casper-md5check.json
/var/log/installer/subiquity-server-debug.log.1693
/var/log/installer/cloud-init.log
/var/log/installer/installer-journal.txt
/var/log/installer/media-info
/var/log/installer/autoinstall-user-data
/var/log/installer/subiquity-client-info.log.1675
/var/log/dpkg.log.1
/var/log/apt/term.log.1.gz
/var/log/apt/history.log
/var/log/apt/term.log
/var/log/apt/history.log.1.gz
/var/log/apt/eipp.log.xz
/var/log/syslog.1
/var/log/kern.log.3.gz
/var/log/kern.log.1
/var/log/journal/9247aebe4e4a4ab9865f790558f3c0b2/user-1000@a9fc17e84b7244169dcc4f0249e664ef-0000000000000ffe-000645bcdcb7bf92.journal
/var/log/journal/9247aebe4e4a4ab9865f790558f3c0b2/user-1000@e2af7c297f2d4c88a6213b5a36c207c4-00000000000004f9-00063f7a698ca2a3.journal
/var/log/journal/9247aebe4e4a4ab9865f790558f3c0b2/system.journal
/var/log/journal/9247aebe4e4a4ab9865f790558f3c0b2/system@476e7ffea4934a2080ed3bad1d192385-0000000000000001-00063f7a6583906a.journal
/var/log/journal/9247aebe4e4a4ab9865f790558f3c0b2/system@476e7ffea4934a2080ed3bad1d192385-0000000000000e88-000645bcd7d1db5a.journal
/var/log/journal/9247aebe4e4a4ab9865f790558f3c0b2/user-1000.journal
/var/log/btmp
/var/log/dmesg.0
/var/snap/lxd/common/config
root@serverubuntu:/home#
```

19. Obtener el UID del usuario root a partir del fichero /etc/passwd

Uso el comando → grep '^root:' /etc/passwd | cut -d: -f3

```
root@serverubuntu:/home# grep '^root:' /etc/passwd | cut -d: -f3
```

```
root@serverubuntu:/home# grep '^root:' /etc/passwd | cut -d: -f3
0
root@serverubuntu:/home#
```

20. Mostrar los usuarios del sistema que tienen un shell válido (excluyendo /usr/sbin/nologin y /bin/false)

Uso el comando → grep -vE '/(nologin|false)$' /etc/passwd | cut -d: -f1

```
root@serverubuntu:/home# grep -vE '/(nologin|false)$' /etc/passwd | cut -d: -f1
```

```
root@serverubuntu:/home# grep -vE '/(nologin|false)$' /etc/passwd | cut -d: -f1
root
sync
linuxserver
root@serverubuntu:/home#
```